

Machine Learning – Regression:

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same., "insurance_pre.csv".

1. **Stage 1:** As the inputs in the datasets are numbers, have chosen Machine Learning Domain.
Stage 2: As the requirements are clear and we have the output in the dataset, supervised learning method is used.
Stage 3: As the output to be predicted is numerical values, it is obvious that we need to create the model using Regression technique/algorithm.
2. The dataset has **1338 rows × 8 columns**.
3. Here, in the dataset we have two categorical columns," Sex" and "Smoker", which we have to convert to numerical columns. For this pre-processing of data, we use **get_dummies** function from pandas library to convert them to numerical values, before splitting the input and output columns. Here, as we have nominal values, **One-hot Encoding** technique is used for conversion.

Also, for the SVM algorithm, for standardizing the inputs, **standardization** method is used before model creation after train and test split of the dataset, so that the inputs will be in a defined range, easier to train the model.

4. First created model using **Multiple Linear Regression** Algorithm, where R2_score is **0.7894**, as the accuracy is **78%**.
5. Further trained the model using **Support Vector Machine** Algorithm, with hyper-tuning parameters, to check for the greater R_score value, which is listed below.

Hyper tuning Parameters	Kernel=Linear (R2_score)	Kernel=rbf (Radial Bias Function) (R2_score)	Kernel=Poly (R2_score)	Kernel=Sigmoid (R2_score)
C=10	0.5665	-0.0181	0.1593	0.0730
C=100	0.6359	0.3906	0.7508	0.5275
C=500	0.7651	0.6964	0.8593	0.4906
C=1000	0.7440	0.8283	0.8605	0.1437
C=2000	0.7414	0.8607	0.8601	-2.5840
C=3000	0.7414	0.8685	0.8600	-6.8261

From the above table, the model performed well for **C=3000** and kernel =”**rbf**”(non-linear), where the accuracy is **86%**.

Next, created model using **Decision Tree** algorithm with the parameter as follows:

S.No.	Criterion	Splitter	R2_score
1.	squared_error	best	0.6908
2.	squared_error	random	0.6637
3.	absolute_error	best	0.6697
4.	absolute_error	random	0.7235
5.	poisson	best	0.7271
6.	poisson	random	0.7569

From the above table, the criterion= “**poisson**” with splitter=” **random**” has the highest R2_score with accuracy of **75%**, which shows that the model did not perform well in this algorithm.

Further created model using **Random Forest** algorithm with the parameter as follows:

S.No.	Criterion	Estimators	R2_score
1.	squared_error	n=10 n=50 n=100	0.8392 0.8514 0.8552
2.	absolute_error	n=10 n=50 n=100	0.8331 0.8543 0.8532
3.	poisson	n=10 n=50 n=100	0.8325 0.8494 0.8535

From the above table, the criterion=" **squared_error**" with estimator value "**n=100**" has the highest R2_score with accuracy of **85%**.

- Overall while comparing the algorithms Multiple Linear Regression, Support Vector Machine, Decision Tree and Random Forest for the dataset, SVM algorithm created the model with high accuracy score. Hence SVM model created is saved in a file with extension .sav.

The saved model is used by the client/end user to load the model and predict the insurance charges for future use.